



CROP2GO

AI-Integrated Digital FPO Platform — Full Architecture & Implementation Manual

A comprehensive, end-to-end technical blueprint designed for the Smart India Hackathon (SIH) 2026. This handbook covers the architecture, complete database schemas, 12-step operational workflows, API specifications, dataset origins, and hand-implementation instructions.

Problem Statement: Student Innovation — Enhancing Indian Agriculture & Processing Agriculture Produce

Problem Statement ID: 26193

Theme: Agriculture, FoodTech & Rural Development

Organization: AICTE / MIC-Student Innovation

Target Platform: Web & Mobile-Responsive Enterprise Agri-Suite

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"We are not replacing FPOs; we are upgrading them into data-driven, AI-assisted, full-service farmer organizations — digitally connecting the entire journey of agricultural produce from farm equipment and collection, through aggregation and processing, to transport, sale, and payment."

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1. Executive Summary & System Philosophy

In India, over 86% of farmers are small and marginal (< 2 hectares). Operating individually, they lack bargaining power, face 30–40% post-harvest wastage due to broken cold chains, and suffer exploitation by middlemen. While **Farmer Producer Organizations (FPOs)** were created to aggregate farmers and achieve economies of scale, most FPOs currently rely on paper registers, informal phone bookings, manual weighing disputes, and lack real-time market intelligence.

CROP2GO transforms traditional FPOs into intelligent, transparent digital hubs by connecting:

- **Digital Traceability:** Every kilogram of produce is tracked from individual farmer lot (`TOM-0609-23`) to bulk aggregated Master Lot (`MASTER-TOM-0609`), dispatch, buyer, and payment settlement.
- **AI Grading ("AI Suggests, Human Confirms"):** Computer vision quality screening classifies produce into A, B, C, and Rejected grades to ensure standardized pricing without disputes.
- **Value-Recovery Processing Pathway:** Automatically routes C-grade produce to food processing units (puree, dehydration, sauce) rather than dumping it at distress rates.
- **Resource Sharing:** Uber-style shared booking of expensive farm machinery (tractors, harvesters) and transport (cold-chain trucks, tempos) with multi-farmer load consolidation.

2. The 4 Primary User Personas

Persona	Access Level	Key Capabilities	Value Proposition
1. Standalone Farmer	Lite Mode	Digital crop diary, expense tracking, mandi market prices, weather warnings, FPO search & application.	User acquisition funnel; upgrades individual farmers into the collective ecosystem.
2. FPO Member Farmer	Full Mode	Instant digital receipts, lot progress tracking, equipment booking, transport booking, proportional payouts.	Eliminates fraud; zero tech literacy needed (operates via SMS/simple app).
3. FPO Admin & Worker	Power User	Intake scanning, digital weighing, AI-assisted grading, master lot aggregation, inventory control, AI alerts, buyer orders, fleet management.	Streamlines daily operations into a single operational dashboard.
4. Institutional Buyer	Commercial	Browse verified master lots, view quality grade breakdowns, submit orders, live GPS shipment tracking, digital invoicing.	Direct access to graded, traceable farm produce at predictable volume.

3. The Master 12-Step Lifecycle Workflow

The entire platform operates along 12 continuous steps divided into intake, intelligence, commercial, and logistics:

Step 1: Farmer Arrival & Collection

Farmer arrives at FPO centre. Worker pulls up profile (or quick-registers). Enters crop type, estimated quantity, and harvest date. System auto-generates unique **Lot ID** (e.g. `T0M-0609-23`) and issues an instant digital delivery receipt.

Step 2: Digital Weighing

Produce is placed on the electronic weighing scale. Worker enters verified weight (e.g. 500 kg). Record is permanently locked to the Lot ID with timestamp and operator ID, preventing "Mera 520 kg tha" quantity disputes.

Step 3: AI-Assisted Quality Grading

Produce sample image or crate is analyzed by computer vision. AI suggests grade breakdown (e.g., 56% Grade A, 32% Grade B, 9% Grade C, 3% Rejected). Worker verifies/adjusts. *Key principle: AI suggests, Human confirms.*

Step 4: Master Lot Aggregation (FPO Superpower)

Individual farmer lots are aggregated into bulk Master Lots (e.g., `MASTER-T0M-0609` : 20,000 kg from 100 farmers). Full traceability is preserved so each farmer receives their exact proportional payout.

Step 5: Live Inventory Dashboard

Displays 4 critical metrics: Received, Sold, Processing, and In-Storage. Real-time alerts for aging produce (> 48h) and cold-room capacity thresholds.

Step 6: AI Decision Intelligence

AI does NOT run the FPO; AI provides actionable intelligence. Evaluates buyer payment history, detects unusual rejection rates, flags processing opportunities, and models spoilage risk based on ambient temperature.

Step 7: Buyer Management & Selling

Institutional buyers browse open master lots and submit orders. FPO managers review buyer reliability ratings (average payment days, past defaults) before confirming sales.

Step 8: Value-Recovery Processing Pathway

C-grade produce that would otherwise be discarded or sold at distress prices is dynamically routed to food processing factories (puree, dehydration, sauce), generating extra revenue for farmers.

Step 9: Dispatch & Traceability Tracking

Confirmed orders are assigned vehicles (cold truck, open truck) and drivers. System creates complete audit trail: *Farmer* → *Lot ID* → *Master Lot* → *Dispatch ID* → *Buyer*.

Step 10: Multi-tier Payment Settlement

When buyer pays FPO, the platform automatically calculates each contributing farmer's share using the exact formula: $(\text{Farmer kg} \times \text{Grade Rate}) \div \text{Master Total}$. Generates itemized digital payment receipts.

Step 11: Shared Equipment & Machine Booking

Farmers book expensive machinery (tractors at \$800/hr, harvesters at \$1,200/hr) from FPO fleet, partner operators, or Govt Custom Hiring Centres (CHC) with calendar scheduling.

Step 12: Coordinated Transport Logistics

Enables farm-to-centre pickup and centre-to-buyer long-haul dispatch. Features **Shared Transport** (pooling 3 small farmer loads into 1 mini truck, saving \$400 per farmer) and cold-chain temperature status.

4. Complete Dataset Inventory & Sourcing Guide

To implement CROP2GO by hand, the system requires 16 total datasets categorized into External APIs, ML Training Datasets, and Platform-Generated Data:

#	Dataset Name	Category	Exact Sourcing Origin	Application Purpose
1	Mandi Market Prices	External API	Agmarknet (agmarknet.gov.in) & data.gov.in	Real-time daily modal prices across Indian APMCs
2	Weather & Climate	External API	OpenWeatherMap / IMD (mausam.imd.gov.in)	Weather alerts, harvest planning, spoilage heat risk
3	FPO Registry Directory	External Static	SFAC (sfacindia.com) & NABARD	Search & discovery of nearby registered FPOs
4	Geolocation & Mapping	External API	OpenStreetMap / Leaflet (free) or Google Maps	Routing, transport tracking, cluster optimization
5	Government Schemes	External Static	PM-KISAN, PMFBY (pmfby.gov.in), eNAM	Credit scoring, crop insurance eligibility
6	Crop Agronomy & Shelf-life	External Static	ICAR (icar.org.in) & FAO Crop Calendar	Sowing seasons, optimal storage temp, aging limits
7	Crop Quality Image Bank	ML Training	Kaggle (Fresh vs Rotten) & PlantVillage	Training MobileNet model for Grade A/B/C classification
8	Historical Mandi Prices	ML Training	data.gov.in 10-year historical mandi archives	Time-series seasonal price trend forecasting
9	Farmer Digital Profiles	Platform Generated	User registration in CROP2GO	Identity, landholding, village, mobile verification
10	Lots & Master Lots	Platform Generated	Step 1 (Intake) & Step 4 (Aggregation)	Lot ID, crop, weight, grade breakdown, chain mapping
11	Live Inventory Levels	Platform Generated	Computed from Lots, Dispatches, and Sales	Warehouse stock tracking across all collection points
12	AI Engine Rule Base	Platform Generated	Computed across transactions & storage logs	Aging alerts, buyer reliability scores, waste risk

#	Dataset Name	Category	Exact Sourcing Origin	Application Purpose
13	Buyer Profiles & Ratings	Platform Generated	Buyer portal transactions	Payment latency, purchase volume, fulfillment trust
14	Dispatch & Logistics Logs	Platform Generated	Step 9 & Step 12 transport bookings	Driver details, vehicle capacity, cold chain temp, GPS
15	Payment & Payout Records	Platform Generated	Step 10 payment distribution engine	Proportional farmer calculation, receipts, ledger
16	Equipment Fleet Register	Platform Generated	FPO & partner machinery inventory	Hourly rental rates, availability schedule, utilization

5. End-to-End System Architecture & Tech Stack

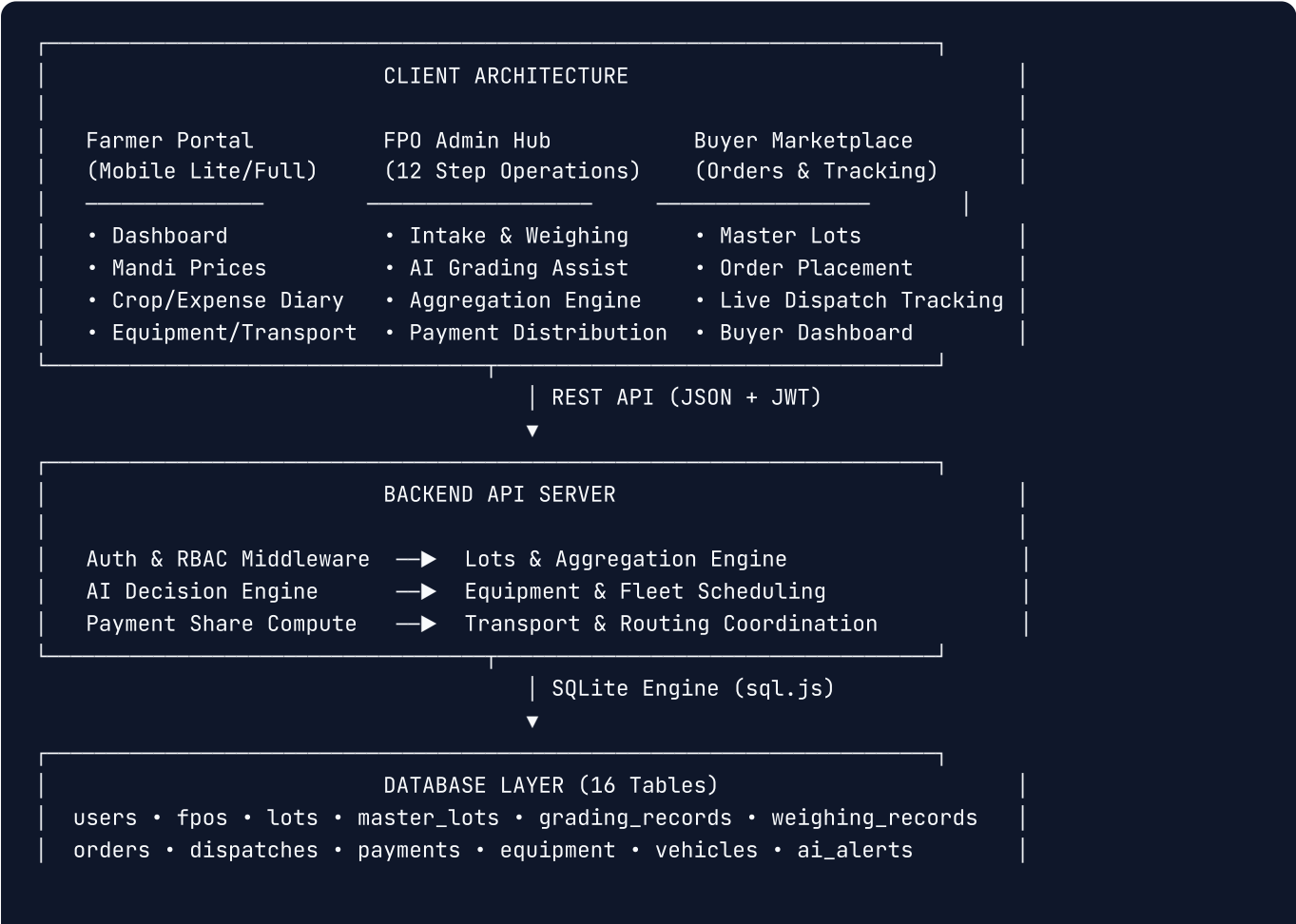
The system is built as a decoupled, high-performance web and mobile-responsive application:

Backend API Server

- **Runtime:** Node.js (v18+) with Express.js
- **Database:** SQLite with `sql.js` wrapper (zero native C++ build dependencies, WAL journal mode, auto-save to disk)
- **Authentication:** JSON Web Tokens (JWT) + bcryptjs salted password hashing
- **Security:** Role-based access control (RBAC) middleware

Frontend Application

- **Framework:** React 18 + Vite (blazing fast builds)
- **Styling:** Tailwind CSS (custom agricultural green/accent palette)
- **Routing:** React Router DOM v6 with role-protected route guards
- **UI Icons:** React Icons (Feather Icon pack)
- **Notifications:** React Hot Toast





6. Complete Database Schema (SQL Definition)

The entire database schema can be executed directly in SQLite / PostgreSQL:

```
-- 1. FPO Organizations
CREATE TABLE fpos (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  name TEXT NOT NULL,
  location TEXT, district TEXT, state TEXT,
  crops_handled TEXT, member_count INTEGER DEFAULT 0, contact_phone TEXT
);

-- 2. Unified User Directory
CREATE TABLE users (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  name TEXT NOT NULL, phone TEXT UNIQUE NOT NULL, password TEXT NOT NULL,
  role TEXT NOT NULL CHECK(role IN ('farmer','fpo_admin','fpo_worker','buyer')),
  fpo_id INTEGER, village TEXT, district TEXT, state TEXT,
  business_name TEXT, business_type TEXT,
  FOREIGN KEY (fpo_id) REFERENCES fpos(id)
);

-- 3. Individual Intake Lots
CREATE TABLE lots (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  lot_id TEXT UNIQUE NOT NULL, farmer_id INTEGER NOT NULL, fpo_id INTEGER NOT NULL,
  crop_type TEXT NOT NULL, variety TEXT, estimated_quantity REAL, weight_kg REAL,
  harvest_date DATE, collection_centre TEXT, recorded_by INTEGER,
  master_lot_id INTEGER, final_grade TEXT, status TEXT DEFAULT 'collected',
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- 4. Weighing & Grading Records
CREATE TABLE weighing_records (
  id INTEGER PRIMARY KEY AUTOINCREMENT, lot_id INTEGER NOT NULL,
  weight_kg REAL NOT NULL, weighed_by INTEGER, weighed_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE grading_records (
  id INTEGER PRIMARY KEY AUTOINCREMENT, lot_id INTEGER NOT NULL,
  grade_a_kg REAL DEFAULT 0, grade_b_kg REAL DEFAULT 0, grade_c_kg REAL DEFAULT 0, rejected_kg REAL,
  ai_suggested_grade TEXT, ai_confidence REAL, final_grade TEXT, graded_by INTEGER
);

-- 5. Master Lots (Aggregation)
CREATE TABLE master_lots (
  id INTEGER PRIMARY KEY AUTOINCREMENT, master_lot_id TEXT UNIQUE NOT NULL,
  fpo_id INTEGER NOT NULL, crop_type TEXT NOT NULL, total_quantity_kg REAL DEFAULT 0,
  grade_a_kg REAL DEFAULT 0, grade_b_kg REAL DEFAULT 0, grade_c_kg REAL DEFAULT 0, rejected_kg REAL,
  contributing_farmers INTEGER DEFAULT 0, status TEXT DEFAULT 'open'
);
```

```
-- 6. Commercial Orders, Dispatches & Payouts
CREATE TABLE orders (
  id INTEGER PRIMARY KEY AUTOINCREMENT, order_id TEXT UNIQUE NOT NULL,
  buyer_id INTEGER NOT NULL, fpo_id INTEGER NOT NULL, master_lot_id INTEGER,
  crop_type TEXT, grade TEXT, quantity_kg REAL, price_per_kg REAL, total_amount REAL,
  status TEXT DEFAULT 'pending'
);

CREATE TABLE dispatches (
  id INTEGER PRIMARY KEY AUTOINCREMENT, dispatch_id TEXT UNIQUE NOT NULL,
  order_id INTEGER, master_lot_id INTEGER, fpo_id INTEGER NOT NULL,
  quantity_kg REAL, destination TEXT, buyer_name TEXT, vehicle_number TEXT,
  vehicle_type TEXT, driver_name TEXT, driver_phone TEXT, cold_chain INTEGER DEFAULT 0,
  status TEXT DEFAULT 'loading', current_location TEXT
);

CREATE TABLE payments (
  id INTEGER PRIMARY KEY AUTOINCREMENT, payment_id TEXT UNIQUE NOT NULL,
  order_id INTEGER, farmer_id INTEGER, fpo_id INTEGER NOT NULL, amount REAL NOT NULL,
  payment_type TEXT DEFAULT 'fpo_to_farmer', grade_breakdown TEXT, status TEXT DEFAULT 'pending',
  paid_at DATETIME
);
```

7. Step-by-Step Hand-Implementation Guide

To reproduce this entire system on any clean computer without AI assistance, follow these exact 6 steps:

Step 1: Install Prerequisites

Ensure Node.js (version 18 or 20+) is installed on your computer. Verify by running:

```
node -v && npm -v
```

Step 2: Create Project Folder Structure

```
mkdir CROP2GO
cd CROP2GO
mkdir -p server/db server/routes server/middleware server/services
mkdir -p client/src/pages/farmer client/src/pages/fpo client/src/pages/buyer
mkdir -p client/src/components client/src/context client/public
```

Step 3: Setup Backend Dependencies

In the `CROP2GO/package.json` file, configure:

```
npm init -y
npm install express cors dotenv jsonwebtoken bcryptjs sql.js
```

Step 4: Setup Frontend Dependencies

Initialize Vite inside the `client` directory:

```
cd client
npm create vite@latest . -- --template react
npm install react-router-dom react-icons axios react-hot-toast chart.js react-chartjs-2
npm install -D tailwindcss postcss autoprefixer
npx tailwindcss init -p
```

Step 5: Populate Database with Realistic Seed Data

Execute the automated seed script to populate all 16 tables with realistic test accounts, active lots, dispatches, and prices:

```
node server/db/seed.js
```

Step 6: Launch Both Server and Client

In Terminal 1 (Backend):

```
node server/index.js
```

In Terminal 2 (Frontend):

```
cd client && npm run dev
```

Open your browser at <http://localhost:5173>.

8. Verification & Live Demo Credentials

Use these pre-seeded demo accounts to test each user role during your hackathon presentation:

Role	Phone Number	Default Password	Demo Script Walkthrough
FPO Admin	9999900001	password123	1. View Live Inventory & AI Alerts. 2. Open Step 1: Collection → Search "Ramesh Patel" → Create Lot. 3. Open Step 2: Weighing → Record weight (500 kg). 4. Open Step 3: Quality Grading → Click "AI Assist" → Confirm Grade. 5. Open Step 4: Aggregation → Group into Master Lot. 6. Open Step 9: Dispatch → Assign Cold Truck → Mark In-Transit. 7. Open Step 10: Payments → Click "Distribute to Farmers".
FPO Farmer	9999900010	password123	1. View dashboard stats, recent crops, and pending payments. 2. Check My Produce to see live grading status. 3. Check Payments for itemized receipt. 4. Book a tractor or shared pickup van in Equipment/Transport.
Standalone Farmer	9999900030	password123	1. Experience Lite Mode (digital diary & expenses). 2. Check live APMC Mandi Rates and IMD Weather warnings. 3. Search nearby FPOs ("Baramati Centre") and click "Apply to Join".
Institutional Buyer	9999900020	password123	1. Browse Available Lots (view Master Lots with grade % bar). 2. Submit order for 8,000 kg A-grade tomato. 3. Monitor delivery progress in Track Shipments with checkpoints.

9. Farmer Section Enhancements & Shared Transport Aggregation

To further empower smallholder farmers and cut logistics overhead, CROP2GO incorporates four major architectural enhancements:

9.1 Cascading Administrative Selection with Password Verification

- **Confirm Password Match:** Real-time matching validation prevents registration typos and blocks form submission on mismatch.
- **Cascading State → District → Village:** Selecting an Indian state dynamically filters agricultural districts and prominent farming villages (with manual custom fallback).

9.2 Complete "My Crop" Lifecycle Tracker

- **Multi-Step Registration Wizard:** Captures field soil type, irrigation baseline, and crop specifications. Supports **"Other / Custom Crop"** to accommodate exotic, spice, or horticultural produce.
- **Crop Growth Timeline & Stages:** Tailored growth models for Wheat, Tomato, Onion, Potato, Rice, Soybean, Cotton, Maize, Mustard, Chilli, and custom crops.
- **AI Harvest Window Prediction:** Calculates early harvest window, most likely harvest date, confidence score, and agronomic reasoning based on soil moisture and variety.
- **Daily Crop Diary:** Interactive stage task checklist, 1-click health status buttons, irrigation toggles, and pest/disease observation logs.
- **Photo + AI Symptom Screening:** Computer-vision screening simulation analyzing crop photos for disease/pest stress with responsible advisory wording (*"Possible issue detected"*).
- **Final Harvest Report:** Side-by-side post-harvest comparison of planned vs actual yield, duration, and AI prediction accuracy.

9.3 Nearby FPO Discovery via Live Geolocation

- **"Use My Live Location" Button:** Uses HTML5 browser geolocation and the mathematical **Haversine formula** to calculate straight-line distances to registered FPOs, sorting closest collection centres first.

9.4 AI-Powered Shared Transport Aggregation

- **Shared Slot Reservation:** Farmers reserve produce slots (kg and target date) rather than renting an entire vehicle individually.
- **Dynamic Demand Consolidation:** Bookings from the same village on the same date automatically aggregate into combined truckloads (e.g. 1,000kg + 500kg + 2,000kg + 1,500kg =

5,000kg total).

- **Automatic Vehicle Tier Assignment:**
 - 100–300 kg → Small Pickup / Magic
 - 301–1,500 kg → Mini Truck
 - 1,501–5,000 kg → Standard 6-Wheeler Truck
 - 5,001+ kg → Heavy Commercial Multi-Truck
- **Economic Benefit:** Reduces transport expenses from ₹3.50/kg to ~₹0.85/kg, delivering **65%–75% cost savings** for smallholder farmers.

SIH Presentation Pro-Tip: Open two browser windows side-by-side: Window 1 as *FPO Admin*, Window 2 as *Farmer*. Show how entering weight and grading in the FPO console instantly updates the farmer's mobile receipt in real-time. This demonstrates complete transparency and zero manual paperwork!